

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56a)

Metric

$$ds^2 = \left(-\frac{\kappa_0 e^x q^2 - 4rm}{2r^2} \right) du \otimes du - du \otimes dr - dr \otimes du + \frac{r^2}{P^2} dx \otimes dx + \frac{r^2}{P^2} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Metric Functions

$$P(x) = P$$

$$q(u) = q$$

$$m(x, u) = m$$

Null Tetrad

Covariant components

$$l = (1) \, du$$

$$n = \left(\frac{\kappa_0 e^x q^2}{4r^2} - \frac{m}{r} \right) du + (1) \, dr$$

$$m = \left(\frac{\sqrt{2}r}{2P} \right) dx + \left(-\frac{i\sqrt{2}r}{2P} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}r}{2P} \right) dx + \left(\frac{i\sqrt{2}r}{2P} \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) du + \left(\frac{\kappa_0 e^x q^2}{4r^2} - \frac{m}{r} \right) dr$$

$$m = \left(\frac{\sqrt{2}P}{2r} \right) dx + \left(-\frac{i\sqrt{2}P}{2r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}P}{2r} \right) dx + \left(\frac{i\sqrt{2}P}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{\kappa_0 e^x q^2}{4r^2} - \frac{m}{r} \partial_r$$

$$\delta = \frac{\sqrt{2}P}{2r} \partial_x - \frac{i\sqrt{2}P}{2r} \partial_y$$

$$\overline{\delta} = \frac{\sqrt{2}P}{2r} \partial_x + \frac{i\sqrt{2}P}{2r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\kappa_0 e^x q^2}{4 r^3} - \frac{m}{r^2}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2} \kappa_0 P e^x q^2}{8 r^3} - \frac{\sqrt{2} P \frac{\partial}{\partial x} m}{2 r^2}$$

$$\alpha = \frac{\sqrt{2} \frac{\partial}{\partial x} P}{4 r}$$

$$\beta = -\frac{\sqrt{2} \frac{\partial}{\partial x} P}{4 r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{\kappa_0 e^x q^2}{4 r^3} - \frac{m}{2 r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 e^x q^2}{4 r^4} - \frac{\frac{\partial}{\partial x} P^2}{4 r^2} + \frac{P \frac{\partial^2}{(\partial x)^2} P}{4 r^2}$$

$$\Phi_{12} = \frac{\sqrt{2} \kappa_0 P e^x q^2}{8 r^4} - \frac{\sqrt{2} P \frac{\partial}{\partial x} m}{4 r^3}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{\sqrt{2} \kappa_0 P e^x q^2}{8 r^4} - \frac{\sqrt{2} P \frac{\partial}{\partial x} m}{4 r^3}$$

$$\Phi_{22} = \frac{\kappa_0 P^2 e^x q^2}{8 r^4} + \frac{\kappa_0 e^x q \frac{\partial}{\partial u} q}{2 r^3} - \frac{P^2 \frac{\partial^2}{(\partial x)^2} m}{2 r^3} - \frac{\frac{\partial}{\partial u} m}{r^2}$$

$$\Lambda = -\frac{\frac{\partial}{\partial x} P^2}{12 r^2} + \frac{P \frac{\partial^2}{(\partial x)^2} P}{12 r^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{\kappa_0 e^x q^2}{2 r^4} + \frac{\frac{\partial}{\partial x} P^2}{6 r^2} - \frac{P \frac{\partial^2}{(\partial x)^2} P}{6 r^2} - \frac{m}{r^3}$$

$$\Psi_3 = \frac{\sqrt{2} \kappa_0 P e^x q^2}{4 r^4} - \frac{3 \sqrt{2} P \frac{\partial}{\partial x} m}{4 r^3}$$

$$\Psi_4 = \frac{\kappa_0 P^2 e^x q^2}{8 r^4} + \frac{\kappa_0 P e^x q^2 \frac{\partial}{\partial x} P}{4 r^4} - \frac{P \frac{\partial}{\partial x} P \frac{\partial}{\partial x} m}{r^3} - \frac{P^2 \frac{\partial^2}{(\partial x)^2} m}{2 r^3}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"